Qualcomm

Qualcomm Technologies, Inc.

Quick Start Guide

HDK8650 (SM8650P) Development Kit

80-61511-1 Rev. AA

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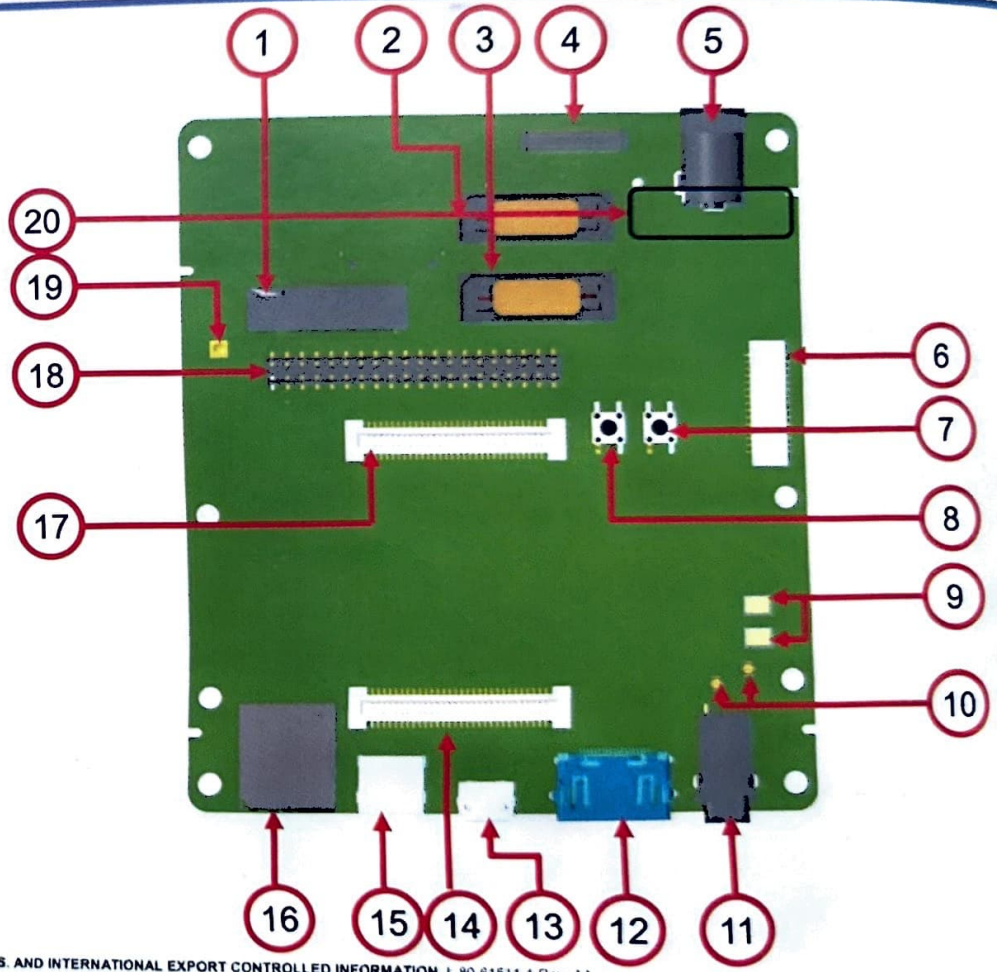
Package Contents

No.	Package item	Optional
1	Main Board with SM8650P processor	
2	12V DC power adaptor	
3	USB cable	
4	Speaker x 2	
5	Display Card	Optional
6	Camera Card	Optional
7	Sensor Card	Optional
8	GNSS Card	Optional
9	Battery	Optional

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Main Board (Top)

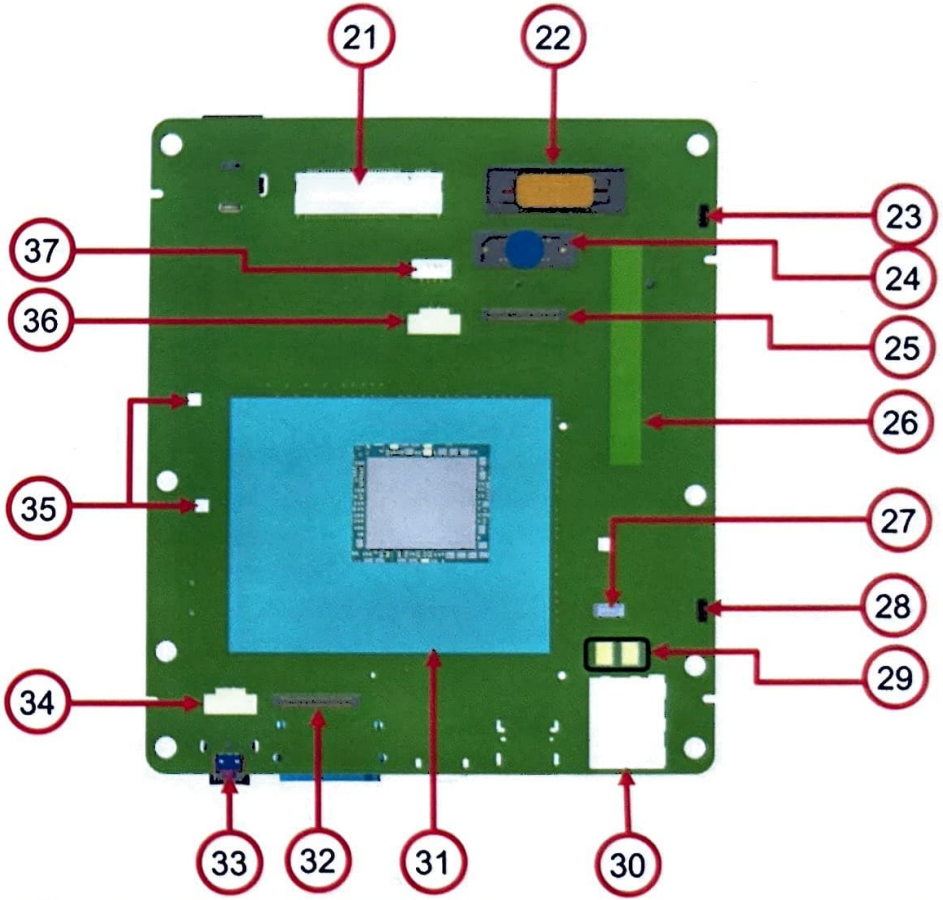
1. M.2 connector
2. Sensor Port2
3. Sensor Port1
4. Camera Card connector1
5. 12V DC Jack
6. Vertigo
7. Volume+ key
8. Volume- key
9. MIC3/4
10. Cable connector(for 3rd party USB repeater)
11. 3.5mm Headset Jack
12. HDMI connector
13. Micro USB
14. High Speed Expansion-1
15. USB Type-C
16. SD Socket
17. High Speed Expansion-2
18. Low Speed Expansion
19. ANT connector
20. User LEDs



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Main Board (Bottom)

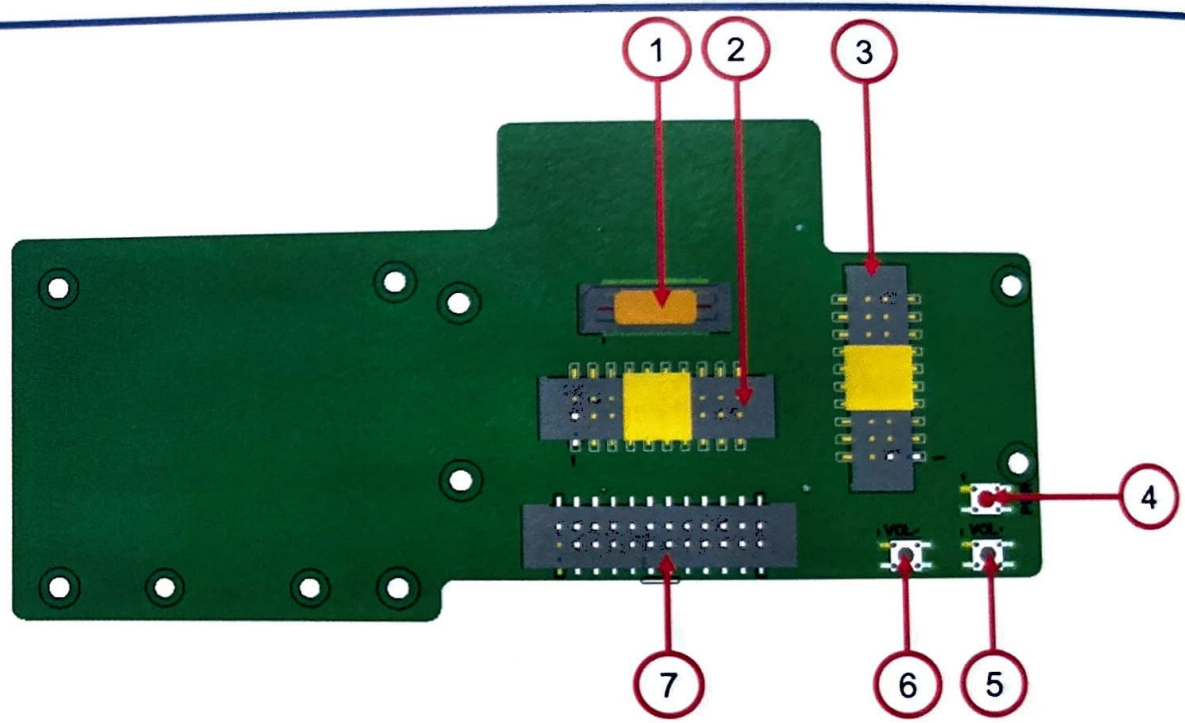
21. M.2 connector
22. NFC/GNSS connector
23. Battery connector1
24. NorFlash connector
25. RCM connector #2
26. Rear Camera connector
27. Fingerprint connector
28. Battery connector2
29. MIC1/5
30. SIM Socket
31. Chip Module
32. Debug/RCM connector #1
33. Power key
34. Speaker connector
35. WIFI ANT connector
36. Receiver connector
37. Fan connector



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Expansion Card (Top)

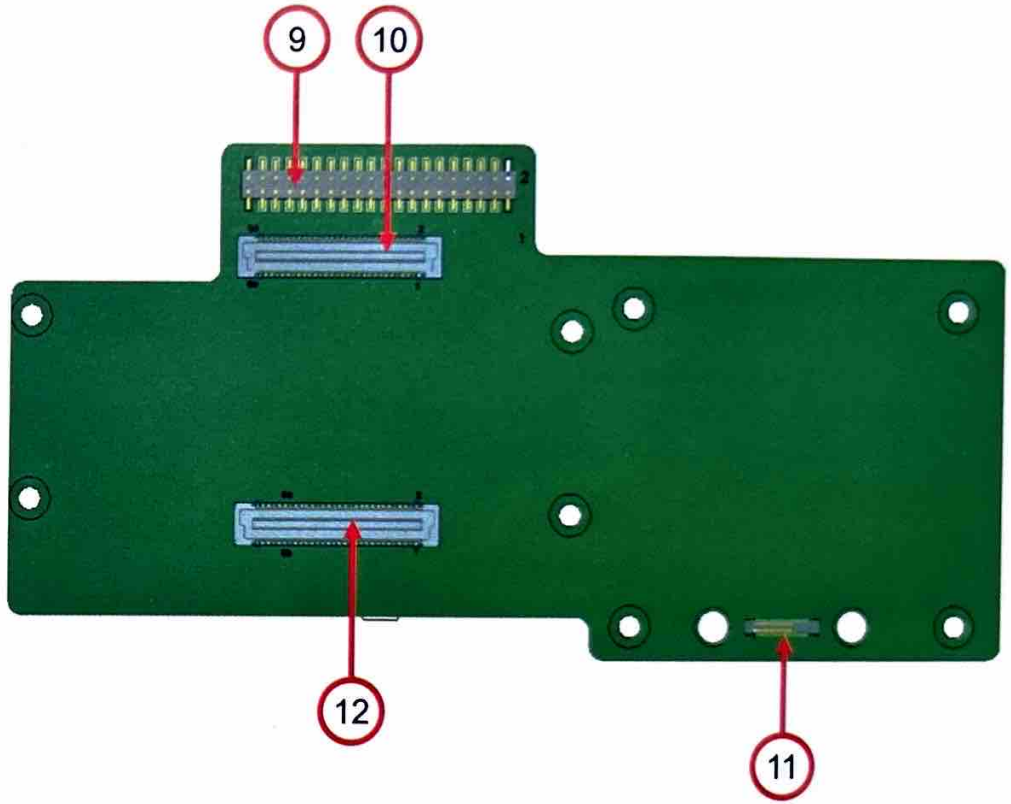
1. Sensor Port3
2. Audio Connector
3. JTAG Connector
4. Power On
5. Volume +
6. Volume -
7. Sensor Port4 (legacy)



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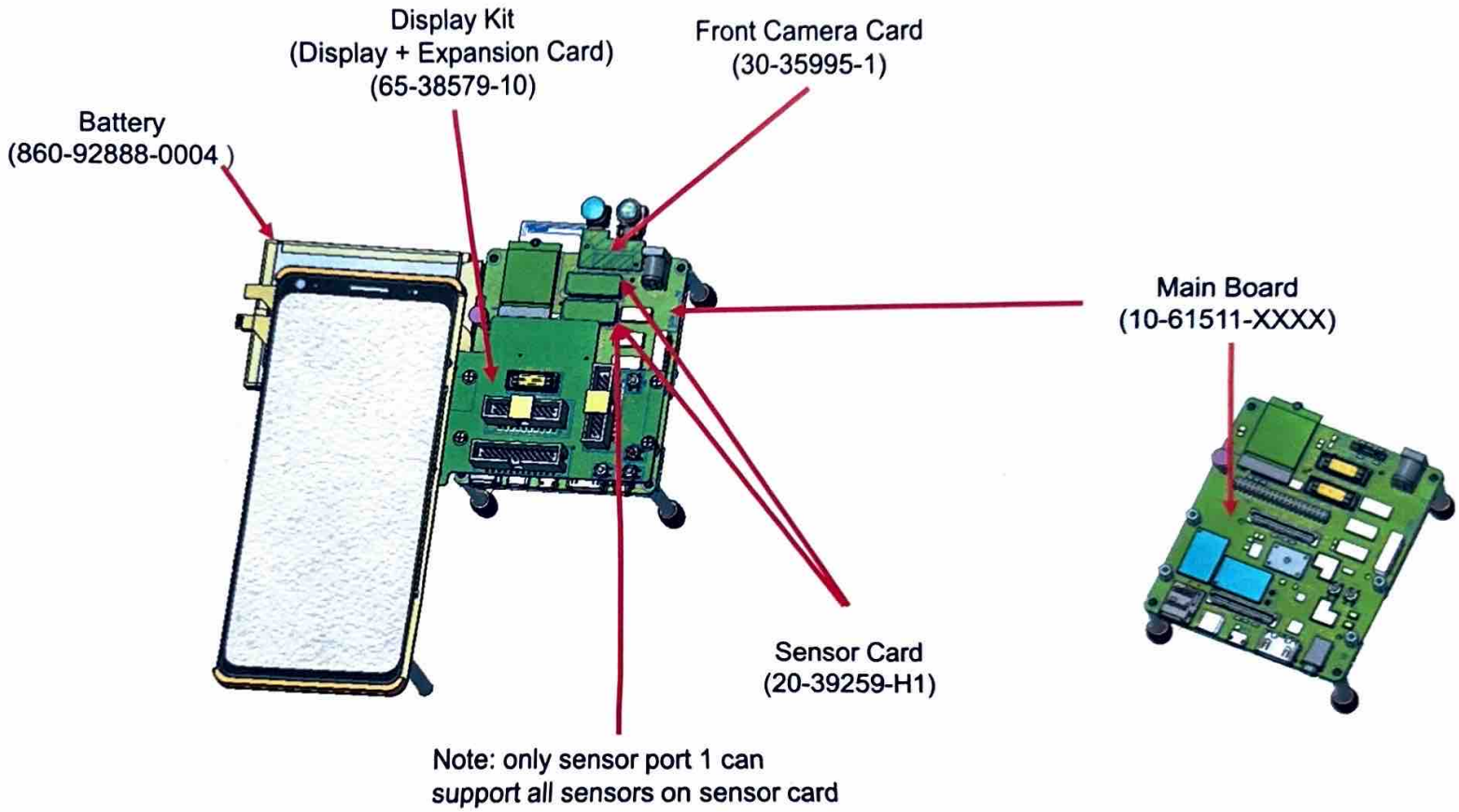
Expansion Card (Bottom)

9. Low Speed Expansion
10. High Speed Expansion-2
11. Display FPC Connector
12. High Speed Expansion



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Development Kit Accessories



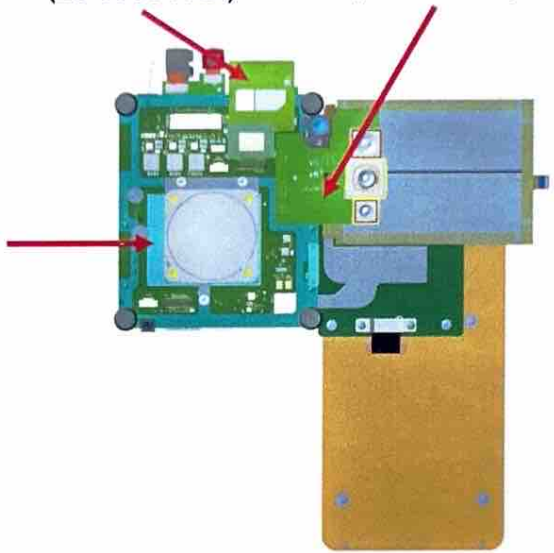
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Development Kit Accessories (cont.)

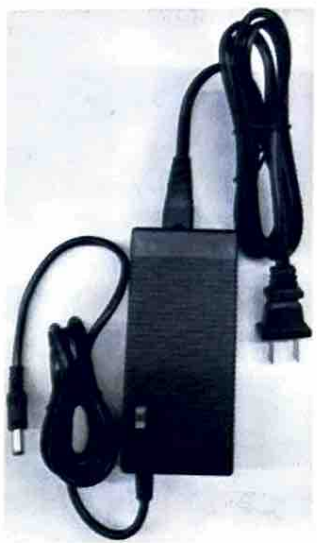
GNSS Card
(20-63361-XX)

Rear Camera Card
(30-28778-1)

LGA Module
(20-62611-XX)



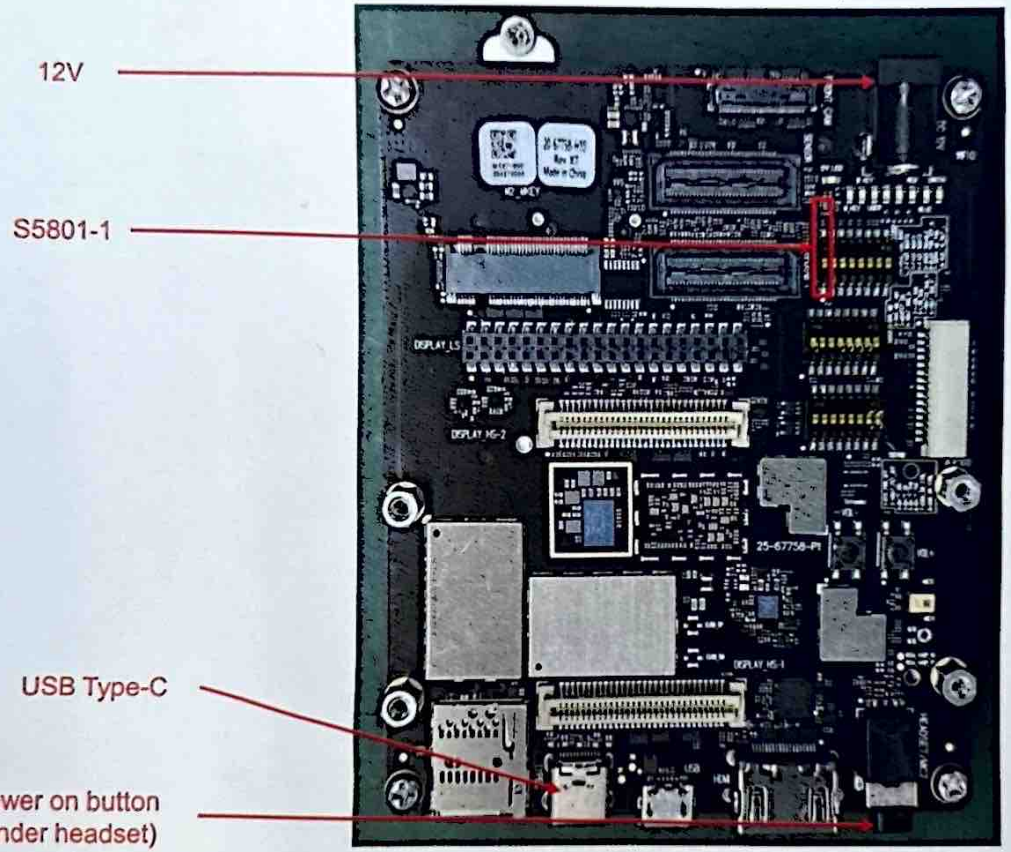
12V DC Adaptor



Notes: Rear camera connector does not support fool-proofing. It causes power issue or even board damage with the wrong installation of this module.

Emergency Download

- 1. Insert the USB Type-C cable.
- 2. Ensure S5801 switch1(Force_usb_boot) is turned **ON** before power-on the device.



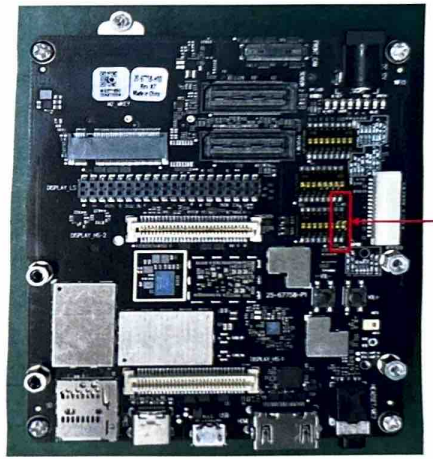
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wer-on the device.



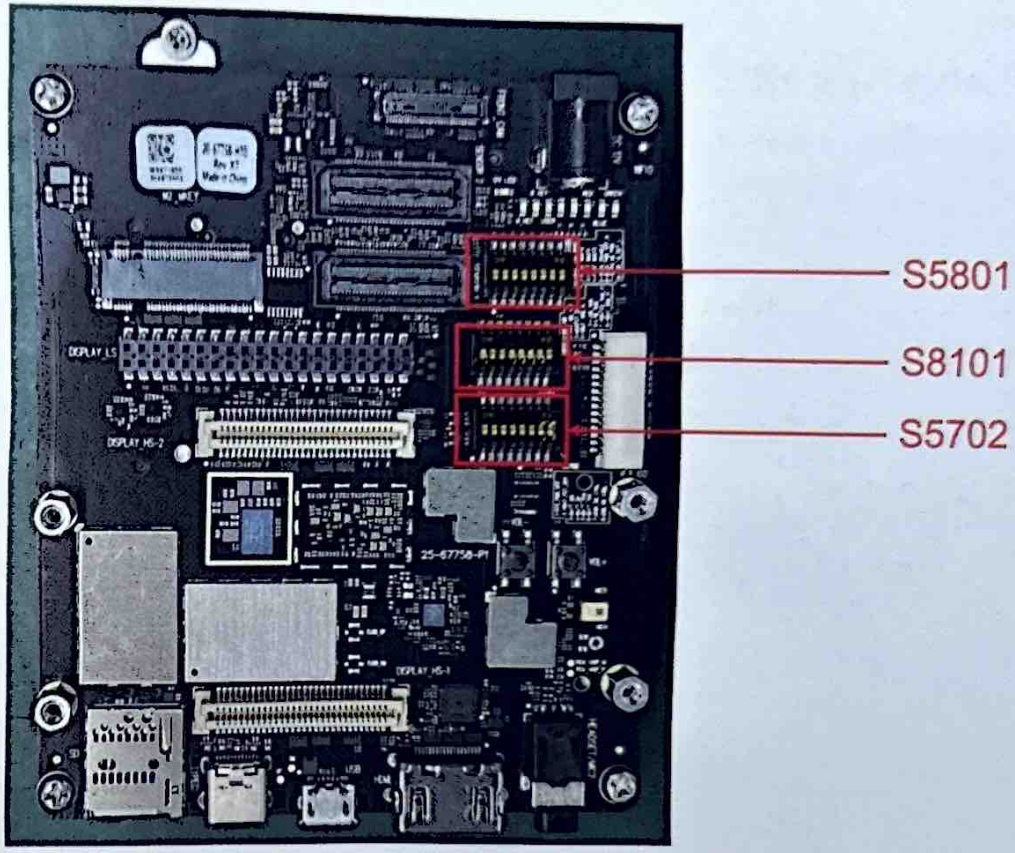
HDMI Function Output

- 1. Insert the HDMI cable.
- 2. Ensure that S5702 switch7 and 8 (DSI0_SW_CTRL, DSI1_SW_CTRL) are turned **ON** before powered on the device.
- 3. Default channel is the display channel.



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DIP Switch Description



DIP Switch Description-S5801

Function	DIP Switch	Description	Notes
FORCED_USB_BOOT	S5801-1	Enable Emergency Download when DIP switch turned on. Controlled by GPIO_83	Default out-of-the-box configuration is OFF.
BOOT_CONFIG(1)	S5801-2	Enable other boot configuration when DIP switch turned on. Controlled by GPIO_144	Default out-of-the-box configuration is OFF.
BOOT_CONFIG(2)	S5801-3	Enable other boot configuration when DIP switch turned on. Controlled by GPIO_145	Default out-of-the-box configuration is OFF
BOOT_CONFIG(3)	S5801-4	Enable other boot configuration when DIP switch turned on. Controlled by GPIO_146	Default out-of-the-box configuration is OFF.
WDG_DISABLE	S5801-5	Enable WATCHDOG_DISABLE when DIP switch turned on. Controlled by GPIO_158	Default out-of-the-box configuration is OFF.
BUCK_PWR_CTRL	S5801-6	Manually disable external 5V(backup channel with 8A output) buck output.	Default out-of-the-box configuration is OFF.
M2_PWR_CTRL	S5801-7	Manually switch 1.8 V and 3.3V power supply for M.2 card	Default out-of-the-box configuration is OFF(3.3V).
HEADSET_DET	S5801-8	Manually connect USB-C headset detect signal to CODEC	Default out-of-the-box configuration is OFF(3.5mm headset with analog audio).

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DIP Switch Description-S5702

Function	DIP Switch	Description	Notes
SYS_PWR_CTRL	S5702-1	Manually disable the external 9 V buck output. Default the buck is ON when DC adapter is plugged in.	Default out of the box configuration is OFF
BATT_FET_CTRL	S5702-2	Manually disable the battery supply path to system. Default the path is controlled by hardware automatically.	Default out of the box configuration is OFF
BATT_ID0	S5702-3	Connect BATT_ID to resistor 1(7.5 K). Default is disconnected.	Default out of the box configuration is OFF
BATT_ID1	S5702-4	Connect BATT_ID to resistor 2(reserved). Default is disconnected.	Default out of the box configuration is OFF
BATT_THERM0	S5702-5	Connect BATT_THERM to resistor 1(100 K). Default is disconnected.	Default out of the box configuration is OFF
BATT_THERM1	S5702-6	Connect BATT_THERM to resistor 1(reserved). Default is disconnected.	Default out of the box configuration is OFF
DSI0_SW	S5702-7	Manually select DSI0 data path between the display panel and HDMI. Default route DSI0 to the display panel.	Default out of the box configuration is OFF
DSI1_SW	S5702-8	Manually select DSI1 data path between the display panel and HDMI. Default route DSI1 to the display panel.	Default out of the box configuration is OFF

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DIP Switch Description-S8101

Function	DIP Switch	Description	Notes
L12B_SW	S8101-5	Connect to the L12B_SNS	Default on On: enable the L12B feedback from HDK main board Off: enable the L12B feedback from display card, can't power on without display card, set to off only if display is abnormal when connecting display card
L11B_SW	S8101-6	Connect to the L11B_SNS	Default on On: enable the L12B feedback from HDK main board Off: enable the L12B feedback from display card, can't power on without display card, set to off only if display is abnormal when connecting display card
VBATT_CONN_VSENSE_MID	S8101-1	Control the VBATT_CONN_VSENSE_MID connection path	Default ON OFF: when using external battery for power supply
	S8101-2		Default OFF ON: when using external battery for power supply
VBATT_CONN_VSENSE_M	S8101-3	Control the VBATT_CONN_VSENSE_M connection path	Default OFF ON: when using external battery for power supply
	S8101-4		Default ON OFF: when using external battery for power supply
VBATT_CONN_VSENSE_P	S8101-7	Control the VBATT_CONN_VSENSE_P connection path	Default ON OFF: when using external battery for power supply
	S8101-8		Default OFF ON: when using external battery for power supply

*12V DC: S8101-1/4/7 ON 2/3/8: OFF
Battery: S8101-1/4/7 OFF 2/3/8: ON